

Prepared by: Syuzanna
Poghosyan, Ekaterina
Shemetova, Anton
Shestakov

Data Analysis with AI: Assignment 3

Research Questions Chosen

How do hotel
occupancy rates vary
across different cities
and months?

What is the relationship
between hotel star
ratings and occupancy
rates?

How does a hotel's
average rating impact
its revenue per room?

What is the effect of
city population size and
tourism ranking on
hotel performance in
that city?

Do hotels with more
amenities experience
higher occupancy rates
or higher reviews
scores?

How do hotel occupancy rates vary across different cities and months?

	month	year	occupancy_rate	city_Graz	city_Hallstatt	city_Linz	city_St. Anton	city_Vienna	city_Zell am See
0	1	2023	0.98	False	False	False	False	True	False
1	1	2024	0.98	False	False	False	False	True	False
2	2	2023	0.98	False	False	False	False	True	False
3	2	2024	0.98	False	False	False	False	True	False
4	3	2023	0.98	False	False	False	False	True	False
...	...	...	...	...	...	...	...	...	...
3031	9	2023	0.98	False	False	False	False	True	False
3032	9	2024	0.98	False	False	False	False	True	False
3033	10	2023	0.98	False	False	False	False	True	False
3034	11	2024	0.98	False	False	False	False	True	False
3035	12	2024	0.98	False	False	False	False	True	False

1782 rows × 9 columns

Example of merged data where city names were transformed into dummies (with true/false)

- 01

Merging datasets

'monthly_occupancy_modified' and hotels_modified' on Hotel_id by left join to keep the main data needed. After that the resulted dataset left join with cities_modified on city.
- 02

Cleaning the data by dropping missing values
- 03

Transforming categorical variables into dummies

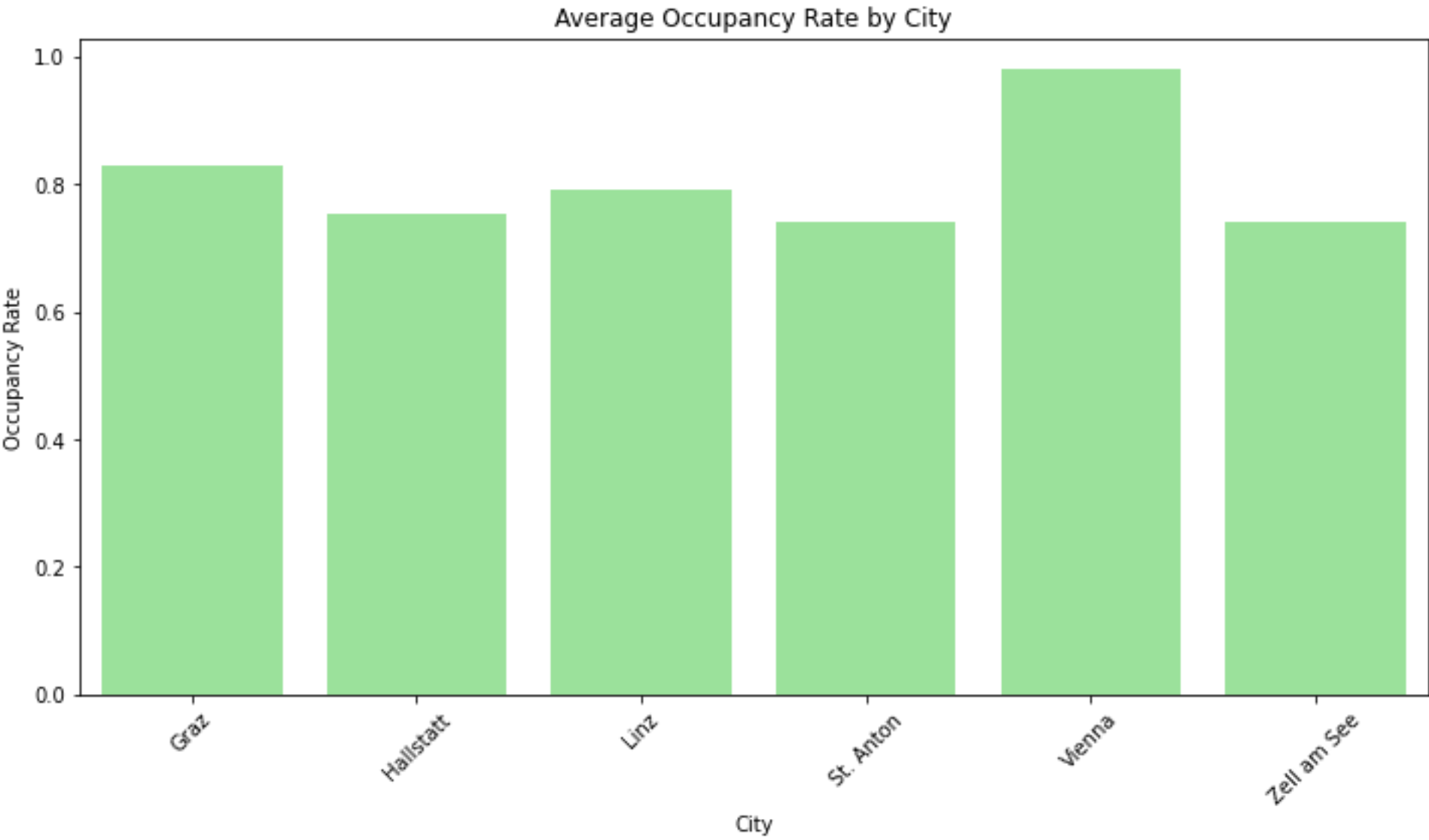
Month or/and city names

By city

OLS Regression Results						
=====						
Dep. Variable:	occupancy_rate	R-squared:	0.867			
Model:	OLS	Adj. R-squared:	0.866			
Method:	Least Squares	F-statistic:	1922.			
Date:	Mon, 24 Mar 2025	Prob (F-statistic):	0.00			
Time:	00:10:36	Log-Likelihood:	3379.6			
No. Observations:	1782	AIC:	-6745.			
Df Residuals:	1775	BIC:	-6707.			
Df Model:	6					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.2733	1.45e+11	-1.88e-12	1.000	-2.85e+11	2.85e+11
city_Graz	1.0832	1.45e+11	7.46e-12	1.000	-2.85e+11	2.85e+11
city_Hallstatt	1.0592	1.45e+11	7.3e-12	1.000	-2.85e+11	2.85e+11
city_Linz	1.0604	1.45e+11	7.31e-12	1.000	-2.85e+11	2.85e+11
city_St. Anton	1.0956	1.45e+11	7.55e-12	1.000	-2.85e+11	2.85e+11
city_Vienna	1.2838	1.45e+11	8.85e-12	1.000	-2.85e+11	2.85e+11
city_Zell am See	1.0560	1.45e+11	7.28e-12	1.000	-2.85e+11	2.85e+11
=====						
Omnibus:	292.642	Durbin-Watson:	0.651			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	536.926			
Skew:	1.016	Prob(JB):	2.56e-117			
Kurtosis:	4.761	Cond. No.	5.27e+14			
=====						

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The smallest eigenvalue is 8.98e-27. This might indicate that there are strong multicollinearity problems or that the design matrix is singular.



Results

The OLS regression aimed to assess variations in occupancy rates across different cities in Austria using dummy variables. The model explains 86.7% of the variance in occupancy rates ($R^2 = 0.867$), but the results indicate severe multicollinearity (smallest eigenvalue = $8.98e-27$). All city coefficients have extremely high standard errors and p-values of 1.000, meaning they are statistically insignificant. This suggests that the model cannot reliably differentiate occupancy rates between cities.

OLS Regression Results						
=====						
Dep. Variable:	occupancy_rate	R-squared:	0.003			
Model:	OLS	Adj. R-squared:	-0.003			
Method:	Least Squares	F-statistic:	0.5614			
Date:	Mon, 24 Mar 2025	Prob (F-statistic):	0.861			
Time:	00:15:58	Log-Likelihood:	1587.9			
No. Observations:	1782	AIC:	-3152.			
Df Residuals:	1770	BIC:	-3086.			
Df Model:	11					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

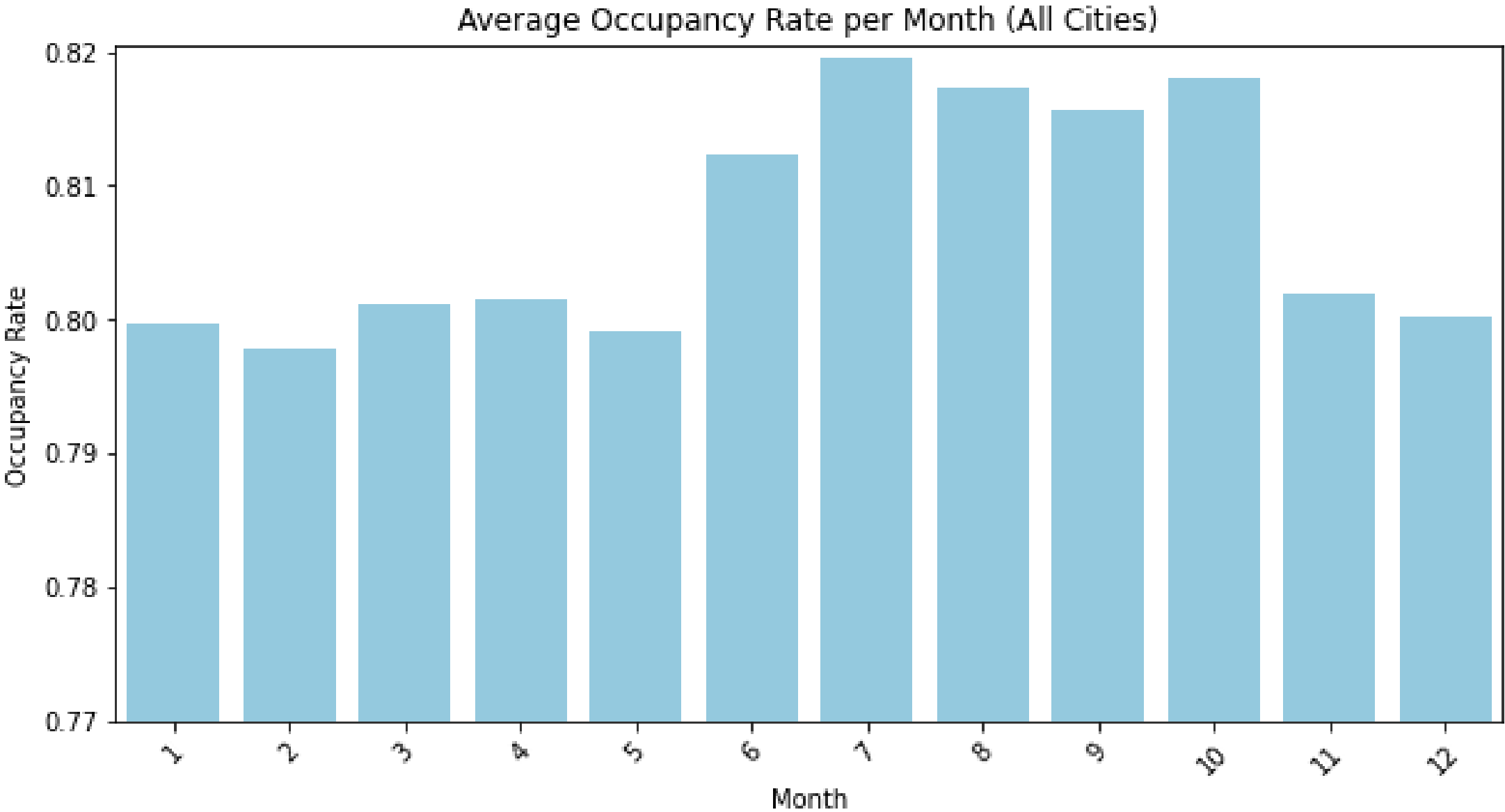
const	0.8257	0.002	379.027	0.000	0.821	0.830
month_1	0.0632	0.008	8.041	0.000	0.048	0.079
month_2	0.0627	0.008	7.928	0.000	0.047	0.078
month_3	0.0647	0.008	8.308	0.000	0.049	0.080
month_4	0.0700	0.008	9.015	0.000	0.055	0.085
month_5	0.0694	0.008	8.930	0.000	0.054	0.085
month_6	0.0675	0.008	8.659	0.000	0.052	0.083
month_7	0.0782	0.008	10.072	0.000	0.063	0.093
month_8	0.0784	0.008	9.813	0.000	0.063	0.094
month_9	0.0762	0.008	9.873	0.000	0.061	0.091
month_10	0.0667	0.008	8.458	0.000	0.051	0.082
month_11	0.0594	0.008	7.510	0.000	0.044	0.075
month_12	0.0691	0.008	8.896	0.000	0.054	0.084
=====						
Omnibus:	16012.778	Durbin-Watson:	0.207			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	219.087			
Skew:	-0.476	Prob(JB):	2.67e-48			
Kurtosis:	1.570	Cond. No.	1.63e+15			
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

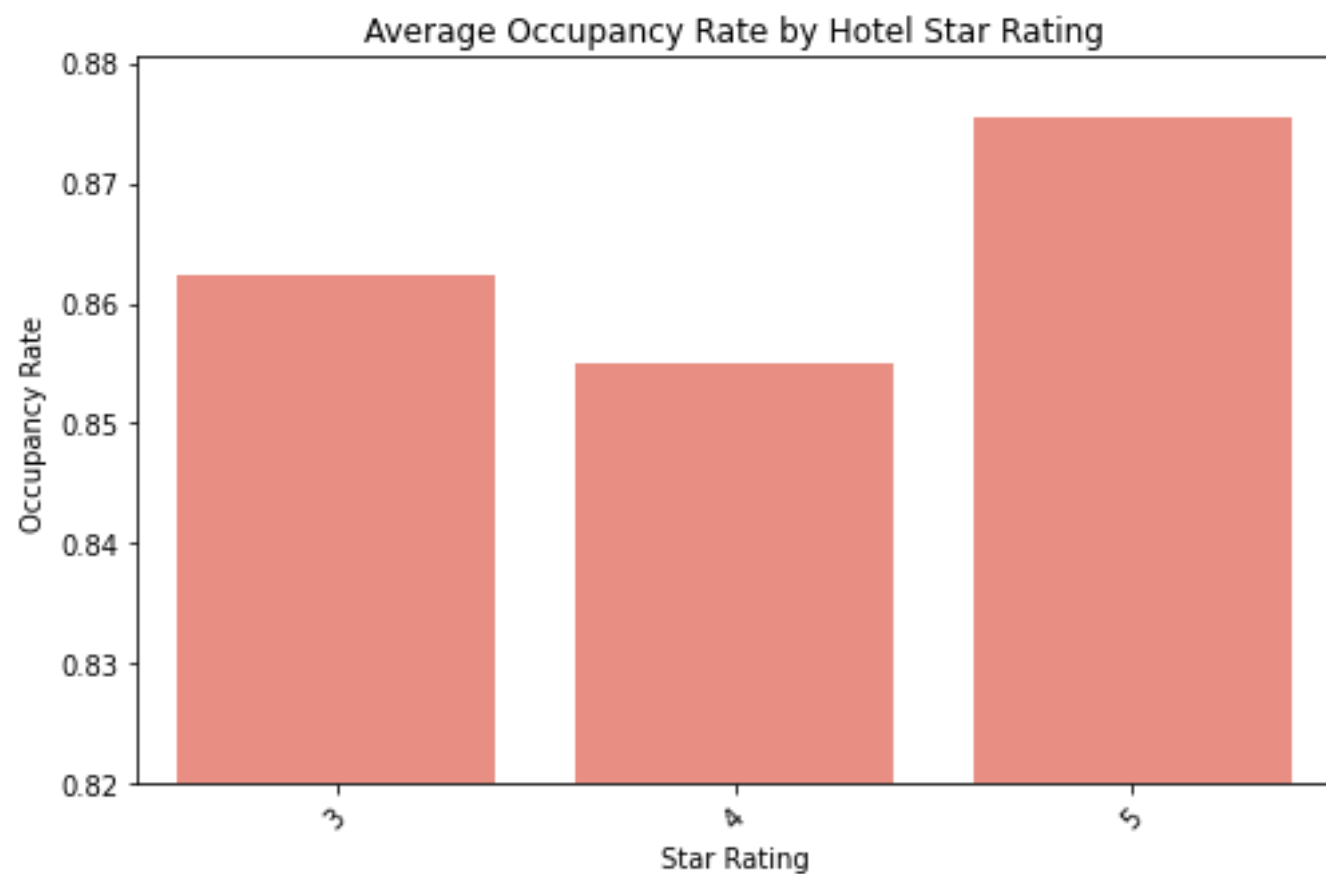
[2] The smallest eigenvalue is 7.24e-28. This might indicate that there are

By month



Results

The regression results indicate that the baseline occupancy rate is 0.8257, with minor variations across months. July (0.0782), August (0.0784), and September (0.0762) show slightly higher occupancy rates, while November (0.0594) has the lowest. Although all months have positive coefficients, the differences are small, and their statistical impact is weak. The model explains only 0.3% of the variance, suggesting that factors other than seasonality play a more significant role in determining occupancy rates.



What is the relationship between hotel star ratings and occupancy rates?

01

Merging data

'hotels_modified' and 'monthly_occupancy_modified' by left merge on 'hotel_id', which will allow to merge the hotels star level and their occupancy ratio.

02

Cleaning the data by dropping missing values

03

Running the OLS-regression

04

Results:

This means that for every 1-unit increase in star_rating, the occupancy_rate increases by 0.65 percentage points, on average.

There is a statistically significant positive relationship between star_rating and occupancy_rate, but the effect size is very small. A higher star rating is associated with a slightly higher occupancy rate, but this relationship explains very little of the overall variation in occupancy rates.

OLS Regression Results

```
=====
Dep. Variable:      occupancy_rate    R-squared:                0.003
Model:              OLS              Adj. R-squared:           0.002
Method:             Least Squares     F-statistic:              6.952
Date:               Mon, 24 Mar 2025   Prob (F-statistic):       0.00843
Time:               00:24:55          Log-Likelihood:          2238.2
No. Observations:   2501             AIC:                     -4472.
Df Residuals:       2499             BIC:                     -4461.
Df Model:           1
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	0.8379	0.010	83.434	0.000	0.818	0.858
star_rating	0.0065	0.002	2.637	0.008	0.002	0.011

```
=====
Omnibus:            13211.106    Durbin-Watson:           0.203
Prob(Omnibus):      0.000       Jarque-Bera (JB):        267.776
Skew:               0.110       Prob(JB):                7.13e-59
Kurtosis:           1.412       Cond. No.                 21.8
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

How does a hotel's average rating impact its revenue per room?

	hotel_id	occupancy_rate	avg_daily_rate	revenue_per_room	rating
0	1	0.980000	206.350000	202.350000	3.075000
1	2	0.832353	271.294118	225.882353	3.880000
2	3	0.805882	243.235294	196.000000	4.080000
3	4	0.822105	220.842105	181.684211	3.466667
4	5	0.838947	253.894737	213.157895	3.925000
...	...	...	...	...	...
137	196	0.808824	173.235294	140.235294	2.862500
138	197	0.815294	153.294118	125.000000	2.600000
139	198	0.792353	169.764706	134.588235	2.978571
140	199	0.980000	206.705882	202.705882	3.060000
141	200	0.815882	262.941176	214.588235	4.007692

- 01

Merging datasets
merging (left type) review and monthly_occupancy datasets after preliminary preparation: we emphasize on finding average values of 'rating' and 'revenue' values neglecting time gap. Eventually, we obtain the following table.
- 02

Cleaning the data by dropping missing values
- 03

Transforming categorical variables into dummies
Logarithmic modification of revenue per room variable due to its right-skewness.

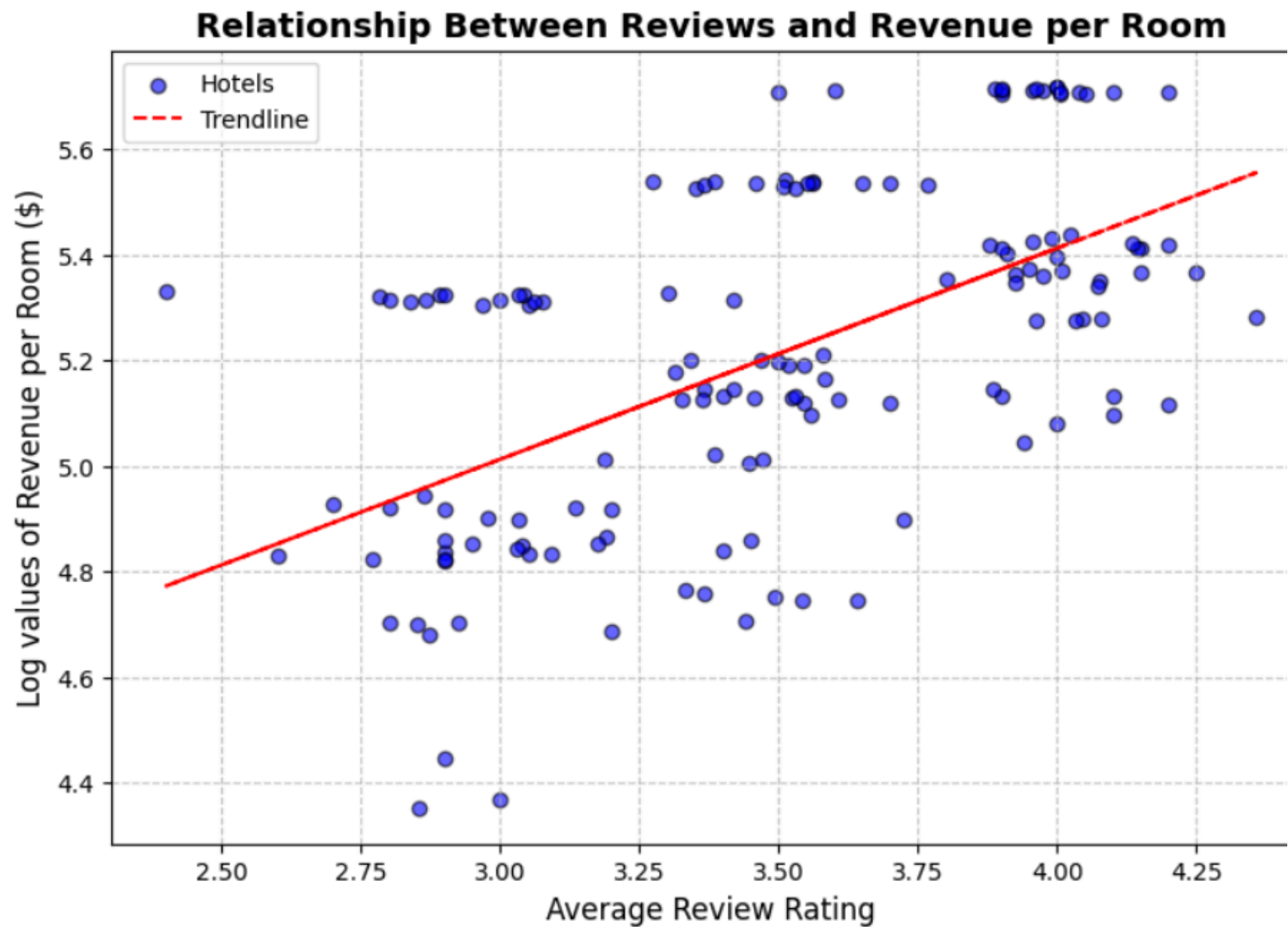


CHART DESCRIPTION

The analysis highlights a clear relationship between hotel reviews with ratings and revenue per room, suggesting that customer satisfaction may play a significant role in financial performance.

While the trendline indicates a positive correlation, other factors such as location, amenities, and pricing strategies could also influence revenue.

Nonetheless, as the first stage of the analysis we are interested in running a simple OLS w/o control vars.

Dep. Variable:	log_revenue_per_room	R-squared (uncentered):	0.989
Model:	OLS	Adj. R-squared (uncentered):	0.989
Method:	Least Squares	F-statistic:	1.233e+04
Date:	Thu, 20 Mar 2025	Prob (F-statistic):	3.79e-139
Time:	23:12:22	Log-Likelihood:	-117.91
No. Observations:	142	AIC:	237.8
Df Residuals:	141	BIC:	240.8
Df Model:	1		
Covariance Type:	nonrobust		
	coef	std err	t P> t [0.025 0.975]
rating	1.4722	0.013	111.057 0.000 1.446 1.498
Omnibus:	1.826	Durbin-Watson:	2.057

OLS RESULTS

The log-transformed OLS regression reveals a strong positive relationship between hotel reviews with ratings and revenue per room, with an extremely high R squared of 0.989, suggesting that ratings explain nearly all the variance in revenue.

The coefficient of 1.4722 ($p < 0.001$) indicates that a one unit increase in rating is associated with a 147.22% increase in revenue per room. The model is highly significant ($F = 12,330$, $p \approx 0$) with well-behaved residuals (Durbin-Watson = 2.057). This suggests that customer ratings are likely a key determinant of hotel revenue

What is the effect of city population size and tourism ranking on hotel performance in that city?

OLS Regression Results						
Dep. Variable:	occupancy_rate	R-squared:	0.948			
Model:	OLS	Adj. R-squared:	0.948			
Method:	Least Squares	F-statistic:	3.584e+05			
Date:	Mon, 24 Mar 2025	Prob (F-statistic):	0.00			
Time:	17:06:16	Log-Likelihood:	92872.			
No. Observations:	39204	AIC:	-1.857e+05			
Df Residuals:	39201	BIC:	-1.857e+05			
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.8526	0.001	1045.879	0.000	0.851	0.854
population	7.464e-08	3.77e-10	197.957	0.000	7.39e-08	7.54e-08
tourism_rank	-0.0141	0.000	-106.896	0.000	-0.014	-0.014
Omnibus:	2025.843	Durbin-Watson:	1.621			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	7792.836			
Skew:	0.069	Prob(JB):	0.00			
Kurtosis:	5.180	Cond. No.	1.02e+07			

OLS Regression Results						
Dep. Variable:	revenue_per_room	R-squared:	0.671			
Model:	OLS	Adj. R-squared:	0.671			
Method:	Least Squares	F-statistic:	3.993e+04			
Date:	Mon, 24 Mar 2025	Prob (F-statistic):	0.00			
Time:	17:06:16	Log-Likelihood:	-1.9782e+05			
No. Observations:	39204	AIC:	3.956e+05			
Df Residuals:	39201	BIC:	3.957e+05			
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	192.7188	1.354	142.376	0.000	190.066	195.372
population	3.684e-05	6.26e-07	58.843	0.000	3.56e-05	3.81e-05
tourism_rank	-9.4529	0.220	-43.028	0.000	-9.884	-9.022
Omnibus:	43124.588	Durbin-Watson:	0.076			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	2439.998			
Skew:	0.006	Prob(JB):	0.00			
Kurtosis:	1.778	Cond. No.	1.02e+07			

performance in that city?

01

Merging data

'cities_modified' and 'cities_tourism_modified' left-merged on City, and the result left-merged initially with 'hotels_modified' by City and then with 'monthly_occupancy_modified' by Hotel_id

02

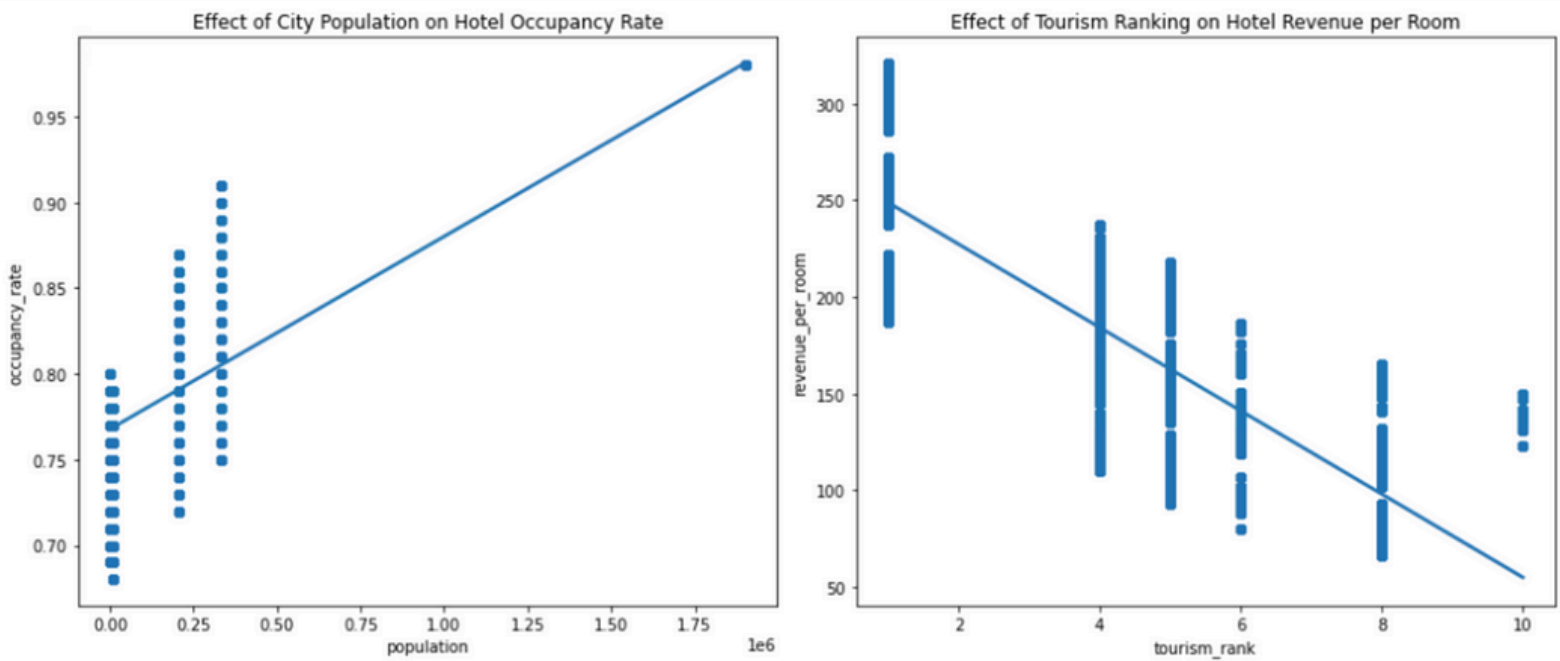
Cleaning the data by dropping the missing values

03

Plotting the relationship of City population on Hotel occupancy rate & Tourism Ranking on Revenue per room only not to overcomplicate everything

04

Running the OLS-regressions to get the relationships



Summary of the Results

Relationship with Occupancy Rate:

- Population size has a very small positive but statistically significant relationship with hotel occupancy rates.
- Higher tourism ranking (lower rank number) - negative and statistically significant correlation with occupancy rate, likely because high-ranking tourism cities have higher supply of hotels.

Relationship with Revenue per Room:

- Population size - a very small positive and statistically significant relationship with revenue per room.
- Higher tourism ranking - negative and statistically significant correlation with revenue per room, meaning hotels in top tourist cities may face competitive pricing.

Do hotels with more amenities experience higher occupancy rates or higher reviews scores?

OLS Regression Results						
Dep. Variable:	occupancy_rate	R-squared:	0.030			
Model:	OLS	Adj. R-squared:	0.030			
Method:	Least Squares	F-statistic:	510.1			
Date:	Mon, 24 Mar 2025	Prob (F-statistic):	3.05e-111			
Time:	16:31:39	Log-Likelihood:	14913.			
No. Observations:	16243	AIC:	-2.982e+04			
Df Residuals:	16241	BIC:	-2.981e+04			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	0.8223	0.002	387.559	0.000	0.818	0.826
num_amenities	0.0090	0.000	22.586	0.000	0.008	0.010
Omnibus:	88300.478	Durbin-Watson:	0.030			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	1655.803			
Skew:	0.116	Prob(JB):	0.00			
Kurtosis:	1.453	Cond. No.	15.3			

OLS Regression Results						
Dep. Variable:	rating	R-squared:	0.403			
Model:	OLS	Adj. R-squared:	0.402			
Method:	Least Squares	F-statistic:	1.094e+04			
Date:	Mon, 24 Mar 2025	Prob (F-statistic):	0.00			
Time:	16:31:39	Log-Likelihood:	-7885.3			
No. Observations:	16243	AIC:	1.577e+04			
Df Residuals:	16241	BIC:	1.579e+04			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	2.6891	0.009	311.427	0.000	2.672	2.706
num_amenities	0.1704	0.002	104.604	0.000	0.167	0.174
Omnibus:	72.432	Durbin-Watson:	1.188			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	55.114			
Skew:	-0.042	Prob(JB):	1.08e-12			
Kurtosis:	2.727	Cond. No.	15.3			

or higher reviews scores?

01

Merging data (all are left merges)

newly created 'amenities_count' with 'hotels_modified' by Hotel_id, then with 'monthly_occupancy_modified' and 'reviews_modified' by Hotel_id

02

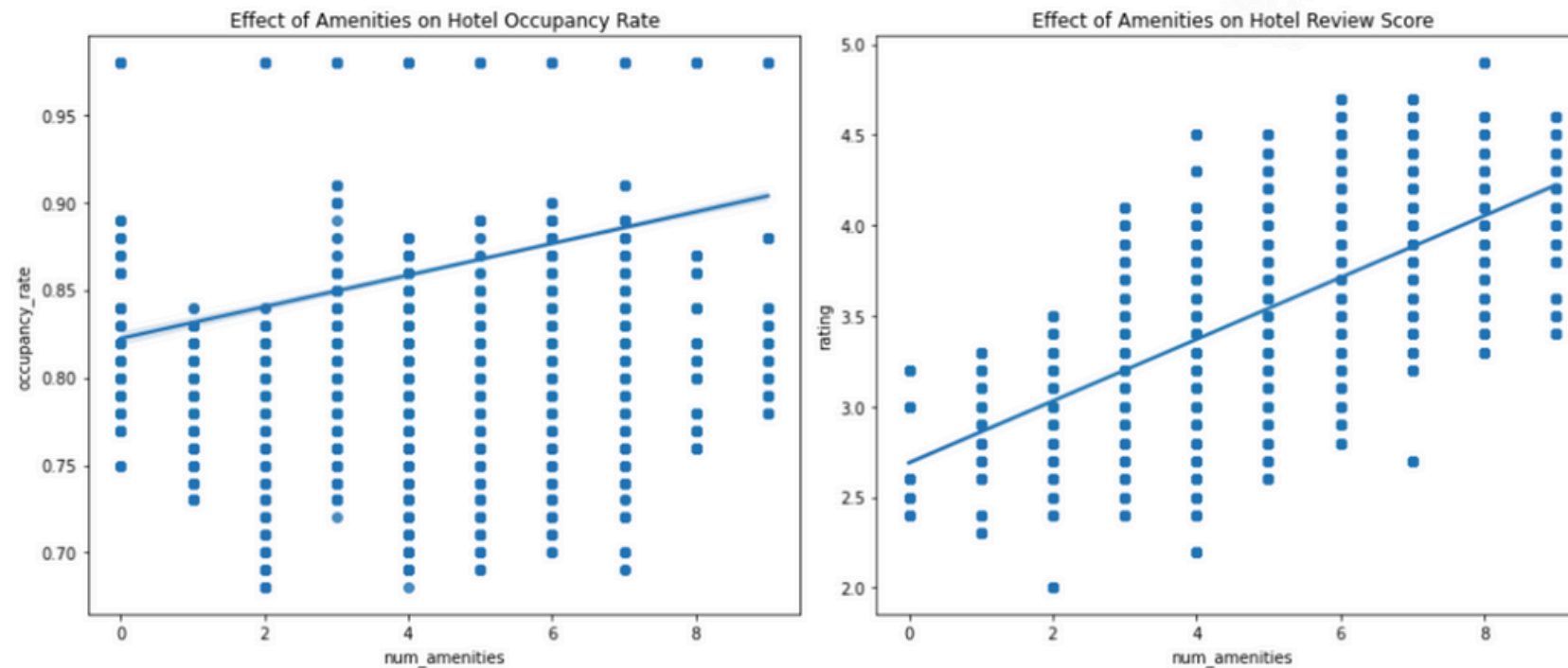
Cleaning the data by dropping the missing values

03

Plotting the relationship of City population on Hotel occupancy rate & Tourism Ranking on Revenue per room

04

Running the OLS-regressions to get the relationships



Summary of the Results

– Amenities vs. Occupancy Rate

The regression output tells us how much occupancy rate changes when amenities increase. As we can see, the coefficient for num_amenities is positive and statistically significant (p-value < 0.05), which suggests that having more amenities is linked to higher occupancy rates.

– Amenities vs. Review Score

Similar logic applies to review scores. As the coefficient for num_amenities is positive and statistically significant, it suggests that hotels with more amenities tend to receive better reviews.